Logical Problem Solving (Week 15, Class 1)

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ANNOUNCEMENT

Presentation

Deliverable 2: 23rd April, at 11:59 pm

What We'll DO today

- ❑ Quiz
- ❑ Predicate Logic

QUIZ

Quiz

- Test will be multiple-choice questions (25 mins; 19 questions)
- Please choose only **one** answer! Circle the correct answer.
- #10, #14, #15, #16: You have to use the eighteen rules of inference to derive the conclusion.
- It is a standard multiple-choice. If you answer correctly, you receive one point. If you get it wrong, you get no point.
- Each correct answer is worth 0.25 points
  - 19 correct answers = 5 points

ANNOUNCEMENT

Quiz 6

April 29th (next Wednesday)

Content: Predicate Logic

- Saturday, May 9th from 08:30 am to 09:50 am in C-109

FINAL EXAM

- **Propositional Logic:**
  - Symbols and Translation
  - Truth Functions
  - Truth Tables for Propositions
  - Truth Tables for Arguments
  - Argument Forms and Fallacies
- **Natural Deduction in Propositional Logic:**
  - Rules of Implication
  - Rules of Replacement
- **Predicate Logic:**
  - Symbols & Translation
  - Using the Rules of Inference

PREDICATE LOGIC

INTRODUCTION

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- **Predicate logic** breaks a proposition (statement) down into a **subject** and a **predicate**.
  - The **subject** is what the statement is about;
  - The **predicate** is what the statement says about the subject.
    - An expression of the form is "… is a bird," "… is a house", and "…are fish"

INTRODUCTION

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- **Predicate logic** breaks a proposition down into a **subject** and a **predicate**.
    - The **subject** is <u>what the statement is about</u>;
    - The **predicate** is what <u>the statement says about the subject</u>.
        - An expression of the form is "… is a bird," "… is a house", and "…are fish"
- The fundamental component in **predicate logic** is the **predicate**.
- **Predicate Logic:**
    - *Def.:* It is a kind of logic that combines the symbolism of *propositional logic* with symbols used to translate ***predicates***.

INTRODUCTION

- Over the remaining two weeks,
  - we'll go over the symbolic notation that is used to represent the inner structure of propositions.
  - Then we'll go on to learn a few new rules of inference and see how to use them (along with the rules in natural deduction we've learned so far) in constructing proofs.
  - Finally, we will see how predicate logic can handle statements about relationships among things

INTRODUCTION

PREDICATE LOGIC

- To understand how to analyse statements in **predicate logic**, we begin by distinguishing the **subject** of a statement from the **predicate**.
  - The **subject** is what the statement is about;
  - The **predicate** is what the statement says about the subject.

PREDICATE LOGIC

- In the following examples, the subject is in **boldface**, and the rest of the sentence is the predicate:

1. **London** is a city.
2. **Jane** got married last Saturday.
3. **Our family** lives in a white house.
4. **Our house** is the last one on the right.
5. **The Civil War** was caused by many factors.
6. Totally unprepared am **I**.
7. **The Milky Way galaxy** is immense.
8. Down the chimney came **Santa**.
9. **Cyprus** was a major source of copper in the Bronze Age.
10. On my left is **Mr. Keith Richards**.

In English, the subject normally comes first, but that is not always the case. In 6, the phrase “totally unprepared” is not the subject; it designates a property of the actual subject, “I”.
In the same way, the subject comes last in
8 and 10.

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Every statement has a subject–predicate structure. But each of our examples below belongs to the subcategory of **singular statements**.

1. **London** is a city.
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PREDICATE LOGIC

- **Singular statement:**
  - makes an assertion about a <u>single</u> thing, as opposed to a general statement about a class of things.
  - The thing can be a person (Jane, Keith Richards), a place (London, Cyprus), an event (the Civil War), or an object (our house).

PREDICATE LOGIC

It can be a group that is treated as a single unit, like "our family" in 3.

3. **Our family** lives in a white house.
4. **Our house** is the last one on the right.
5. **The Civil War** was caused by many factors.
6. Totally unprepared am **I**.
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PREDICATE LOGIC

- In the following examples, the subject is in **boldface**, and the rest of the sentence is the predicate:

Even 7 is a singular statement; although our galaxy contains a great many things, the statement treats it as a single unit, as opposed to a general statement about galaxies (e.g., galaxies tend to have spiral shapes)

6. Totally unprepared am **I**.
7. **The Milky Way galaxy** is immense.
8. Down the chimney came **Santa**.
9. **Cyprus** was a major source of copper in the Bronze Age.
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PREDICATE LOGIC

- The **subject** is symbolized by lowercase letters (a, b, c, ... x, y, z), called **individual constants**.

PREDICATE LOGIC

- In the following examples, the subject is in **boldface**, and the rest of the sentence is the predicate:

1. **London** is a city.
2. **Jane** got married last Saturday.
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	Subject
1.	london
2.	jane
3.	we
4.	our house
5.	the civil war
6.	i
7.	the milky way
8.	cyprus
9.	santa
10.	mr. keith richards

PREDICATE LOGIC

- The **predicate** is symbolized by upercase letters (A, B, C, … X, Y, Z), called **predicate symbols**.
- It is conventional to use capital letters for predicates, lowercase letters for subjects, with the predicate letter given first.
- Here are some examples …

The **subject** is <u>what the statement is about</u>

The **predicate** is what <u>the statement says about the subject</u>.

	Subject	Predicate	Symbolic
1.	**l**ondon	is a **C**ity.	Cl
2.	**j**ane	got **M**arried last Saturday.	Mj
3.	**w**e	**L**ive in a white house.	Lw
4.	our **h**ouse	is the last one on the **R**ight.	Rh
5.	the **c**ivil war	was **C**aused by many factors.	Cc
6.	**i**	am totally **U**nprepared.	Ui
7.	the **m**ilky way	is **I**mmense.	Im
8.	**c**yprus	was a major source of **C**opper in the Bronze Age.	Cc
9.	**s**anta	**C**ame down the chimney.	Cs
10.	mr. **k**eith richards	is on my **L**eft.	Lk

This notation may seem foreign at first, since it reverses the order of English grammar.

It's a bit like Yoda-speak in *Star Wars*

	Subject	Predicate	Symbolic
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2.	**j**ane	got **M**arried last Saturday.	Mj
3.	**w**e	**L**ive in a white house.	Lw
4.	our **h**ouse	is the last one on the **R**ight.	Rh
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7.	the **m**ilky way	is **I**mmense.	Im
8.	**c**yprus	was a major source of **C**opper in the Bronze Age.	Cc
9.	**s**anta	**C**ame down the chimney.	Cs
10.	mr. **k**eith richards	is on my **L**eft.	Lk

It may help to think of the symbolic formula as saying that the predicate is true of the subject: Being a city (C) is true of London (l); getting married last Saturday (M) is true of Jane (j), and so on …

	Subject	Predicate	Symbolic
1.	**l**ondon	is a **C**ity.	Cl
2.	**j**ane	got **M**arried last Saturday.	Mj
3.	**w**e	**L**ive in a white house.	Lw
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PREDICATE LOGIC

1. Translate the following statements into symbolic form

Statement	Symbolic translation
Socrates is mortal.	
Tokyo is populous.	
The *Sun-Times* is a newspaper.	
King Lear is not a fairy tale.	
Berlioz was not a German.	

PREDICATE LOGIC

1. Translate the following statements into symbolic form

Statement	Symbolic translation
Socrates is mortal.	*Ms*
Tokyo is populous.	*Pt*
The *Sun-Times* is a newspaper.	*Ns*
King Lear is not a fairy tale.	~*Fk*
Berlioz was not a German.	~*Gb*

PREDICATE LOGIC

- As in propositional logic, singular statements can be combined into compound statements by means of logical operators …

PREDICATE LOGIC

	Statement	Symbolic
11.	If London is a city, it has a mayor.	$Cl \supset Ml$
12.	Scotland is not a city.	$\sim Cs$
13.	If Cyprus was a major source of copper in the Bronze Age, then Egypt imported copper.	$Cc \supset Ie$
14.	Jane got married last Saturday and so did Tamara.	$Mj \bullet Mt$
15.	Harry's car is an orange convertible.	$Oc \bullet Cc$
16.	I will be either at home or in New York.	$Hi \vee Ni$
17.	If the moon is out tonight and the sea is calm, I'll go for a sail or a walk on the beach.	$(Om \bullet Cs) \supset (Si \vee Wi)$

Note that some compound statements may have the same subject, as in 11, 15, and 16.

	Statement	Symbolic
11.	If London is a city, it has a mayor.	Cl ⊃ Ml
12.	Scotland is not a city.	~Cs
13.	If Cyprus was a major source of copper in the Bronze Age, then Egypt imported copper.	Cc ⊃ Ie
14.	Jane got married last Saturday and so did Tamara.	Mj • Mt
15.	Harry's car is an orange convertible.	Oc • Cc
16.	I will be either at home or in New York.	Hi ∨ Ni
17.	If the moon is out tonight and the sea is calm, I'll go for a sail or a walk on the beach.	(Om • Cs) ⊃ (Si ∨ Wi)

Note that some compound statements may also have the same predicate, as in 14, or differ in both subject and predicate, as in 13.

	Statement	Symbolic
11.	If London is a city, it has a mayor.	$Cl \supset Ml$
12.	Scotland is not a city.	$\sim Cs$
13.	If Cyprus was a major source of copper in the Bronze Age, then Egypt imported copper.	$Cc \supset Ie$
14.	Jane got married last Saturday and so did Tamara.	$Mj \bullet Mt$
15.	Harry's car is an orange convertible.	$Oc \bullet Cc$
16.	I will be either at home or in New York.	$Hi \vee Ni$
17.	If the moon is out tonight and the sea is calm, I'll go for a sail or a walk on the beach.	$(Om \bullet Cs) \supset (Si \vee Wi)$

In 15, we could have treated “orange convertible” as a single predicate, but since the colour and the type of car are independent characteristics, we can also make them separate predicates ascribed to the car.

11.	If London is a city, it has a mayor.	$Cl \supset Ml$
12.	Scotland is not a city.	$\sim Cs$
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14.	Jane got married last Saturday and so did Tamara.	$Mj \bullet Mt$
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PREDICATE LOGIC

- (2) Translate the followings statements into symbolic form

Statement	Symbolic translation
If Paris is beautiful, then Andre told the truth.	
Irene is either a doctor or a lawyer.	
Senator Wilkins will be elected only if he campaigns.	
General Motors will prosper if either Nissan is crippled by a strike or Subaru declares bankruptcy.	
Indianapolis gets rain if and only if Chicago and Milwaukee get snow.	

PREDICATE LOGIC

- (2) Translate the followings statements into symbolic form

Statement	**Symbolic translation**
If Paris is beautiful, then Andre told the truth.	$Bp \supset Ta$
Irene is either a doctor or a lawyer.	$Di \vee Li$
Senator Wilkins will be elected only if he campaigns.	$Ew \supset Cw$
General Motors will prosper if either Nissan is crippled by a strike or Subaru declares bankruptcy.	$(Cn \vee Ds) \supset Pg$
Indianapolis gets rain if and only if Chicago and Milwaukee get snow.	$Ri \equiv (Sc \bullet Sm)$

Predicate Logic

(3) Put each of the following statements into symbolic notation.

1. Tobias is bored.
2. London Bridge is falling down.
3. Randy Newman is not a short guy but Napoleon was.
4. Jim Jones is either crazy or thirsty.
5. Luminous beings are we, not this crude matter. (—Yoda)
6. Macbeth was a tyrant with a conscience.
7. If the streets are clear, I'll drive, but if not, I'll take the train.
8. Whether the baby is a boy or a girl, its name will be Jamie.

PREDICATE LOGIC

1. Bt
2. Fl
3. ~Sr · Sn
4. Cj v Tj
5. Lw · ~Mw
6. Tm · Cm
7. (Cs ⊃ Di) · (~Cs ⊃ Ti)
8. (Bb v Gb) ⊃ Jb